

Tanvi Joshi

Senior Game Engine Programmer (C++)

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SUMMARY

Senior engine programmer with 7 years of hard C++ behind shipped titles: physics simulation, SIMD math libraries, and making frame budgets on fixed hardware. Looking to apply low-level performance engineering to quantitative finance, where the same discipline — fast, correct numerical code under strict budgets — decides outcomes.

Notice period: 45 days.

SKILLS

C++: C++17/20, SIMD (SSE/AVX intrinsics), lock-free structures, custom allocators, profiling (VTune, Tracy), CMake, cross-platform builds

Math: Numerical integration, linear algebra, collision/constraint solvers, floating-point determinism across platforms

Other: Python tooling, Perforce/Git, mentoring (4 juniors)

EXPERIENCE

Senior Engine Programmer — Mysuru Interactive, Bengaluru

Oct 2021 – present

- Own the physics and math libraries for a cross-platform engine used in two shipped titles (4M+ combined downloads).
- Rewrote the constraint solver with AVX2 SIMD: 3.8x throughput, enabling 5x more dynamic objects per scene.
- Achieved cross-platform floating-point determinism for networked physics — the hardest correctness bug hunt of my career.
- Mentor four junior programmers; run the studio's C++ guild.

Gameplay/Engine Programmer — OrangeLoop Games, Pune

Sep 2019 – Sep 2021

- Gameplay systems and engine optimization for mobile titles; hit 60fps targets on constrained hardware.

EDUCATION

B.Tech Computer Engineering — COEP Technological University, Pune, 2015 – 2019

TALKS

"Deterministic floats or bust" — India Game Developer Conference (IGDC), Hyderabad 2024

LANGUAGES KNOWN

Marathi (native), Hindi (fluent), English (fluent), Kannada (basic)